

8) What rules of inference are used in this argument?

"No man is an island. Manhattan is an island. Therefore, Manhattan is not a man."

Soln: $M(x)$: x is a man $I(x)$: x is an island
Domain: Set of all people Domain: Set of all islands.

Premise: 1) $\forall x (M(x) \Rightarrow \neg I(x))$ 2) $I(\text{Manhattan})$

Conclusion: $\neg M(\text{Manhattan})$

Steps	Reason
1) $\forall x (M(x) \Rightarrow \neg I(x))$	Premise
2) $I(\text{Manhattan})$	Premise
3) $M(\text{Manhattan}) \Rightarrow \neg I(\text{Manhattan})$	Universal instantiation on 1
4) $\neg M(\text{Manhattan})$	Modus tollens on 2, 3

9) For each of these collections of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

a) "If I take the day off, it either rains or snows." "I took Tuesday off or I took Thursday off." "It was sunny on Tuesday." "It did not snow on Thursday."

Soln: $O(x)$: I took day x off $R(x)$: It was raining on day x
 $S(x)$: It was snowing on day x $Su(x)$: It was sunny on day x

Domain: Set of all weekdays

Premise: 1) $\forall x (O(x) \Rightarrow R(x) \vee S(x))$ 2) $O(\text{Tuesday}) \vee O(\text{Thursday})$
3) $Su(\text{Tuesday})$ 4) $\neg S(\text{Thursday})$

5) $[Su(\text{Tuesday}) \Rightarrow \neg R(\text{Tuesday}) \wedge \neg S(\text{Tuesday})] \rightarrow \text{Easy to see}$